DriveCompose: Generative Reconstruction of Compositional Driving Scenes via Joint Completion

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Abstract

Driving simulation and scene editing require reconstructed scenes that remain usable when cameras and actors move beyond their recorded configurations. Such changes expose unobserved surfaces, whose geometry and appearance must be completed in a way that preserves their association with the scene and its actors. We present DriveCompose, a framework for generative reconstruction of compositional driving scenes from calibrated image sequences. Our key idea is to jointly complete scene components in the coordinate frames used for their subsequent control. Features recovered from the input images are aggregated onto a world-space background and canonical vehicle and human supports, and a conditional flow model uses cross-component memory to infer their missing latent content from shared scene context. Observed evidence constrains generation, and the assembled scene is checked against the input views before reuse. On Waymo, DriveCompose improves dynamic scene reconstruction and off-trajectory view synthesis over the evaluated baselines. Ablations show that sharing context across components improves completion of unseen surfaces. The resulting Gaussian scene separates content construction from state-driven rendering, keeping the completed background and canonical assets fixed during subsequent camera and actor control.

1. Introduction

Driving datasets record scenes along captured trajectories [1, 31]. Driving simulation and scene editing require a reconstruction that remains usable when the recorded configuration changes. Moving a camera, turning a vehicle, or changing a pedestrian's pose exposes previously unobserved surfaces. These operations require geometry and appearance beyond the recorded images, together with a representation that keeps the content attached to the same scene and actor identities.

Recent feed-forward methods recover driving scenes and motion directly from temporal observations [3, 6, 45, 48]. Recovering observed content alone does not supply the unseen surfaces exposed by later actor motion.

Compositional representations separate background and actors and provide canonical models for independent motion and control [5, 43, 49]. Generative models address missing content through asset priors, Gaussian completion, or generated views [17, 44, 47, 51]. Combining them requires a generator that assigns content to the background and to individual assets. A generated surface must have an owner, a coordinate frame, and a motion rule before it can become part of an editable scene.

We formulate driving-scene reconstruction as conditional generation of a compositional scene. Our representation follows the units that will be manipulated: a world-space background, rigid vehicles in local coordinates, and humans on canonical surfaces with skeleton bindings. Completing surfaces in their component's control frame lets the same generated content retain its identity under later state changes (Sec. 3.1). Image evidence is first aggregated on these supports. A generative model then completes their missing latent content under the observed scene conditions.

DriveCompose implements this formulation with a visual recovery front end, structured latent codes, and a conditional flow model [18] (Fig. 1). The front end estimates geometry, trajectory-level identities, and time-varying states. Within the completion model, local attention follows each component's support, while shared memory exchanges context between background and assets. Type-specific decoders convert the completed codes into Gaussian components.

Generation must also respect the images from which the scene was recovered. We retain reliable observed codes and suppress known free space throughout sampling. Newly generated content can nevertheless occlude an observed surface, so we check the assembled candidate in the original views as well. After construction, the background and canonical assets remain fixed. Camera placement, vehicle transforms, and human joint poses determine their subsequent rendering (Sec. 3.5).

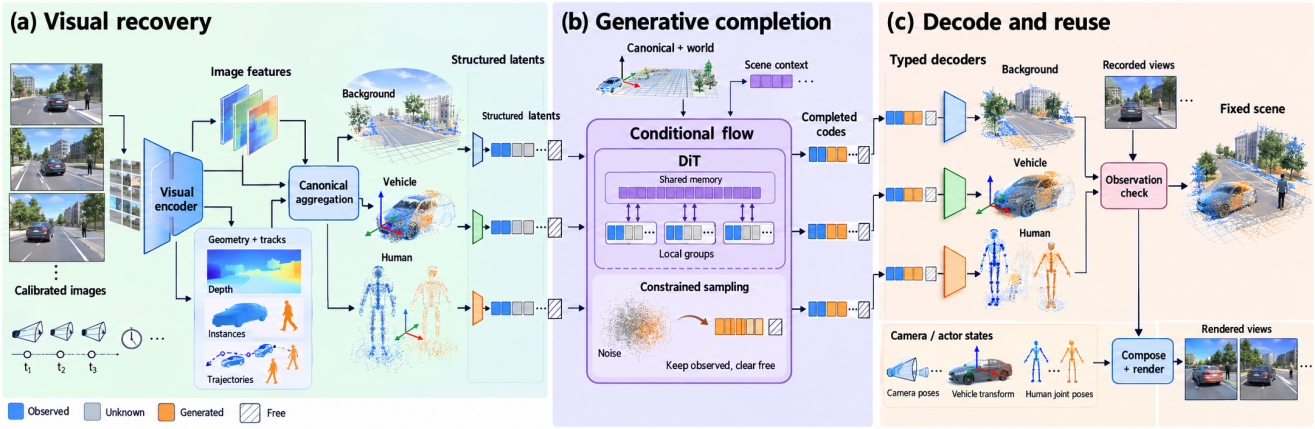


Figure 1. **DriveCompose overview.** (a) Calibrated images provide geometry, identity, and motion evidence, which is aggregated on background and canonical asset supports and encoded into structured latents. (b) A scene-conditioned flow transformer completes unknown codes while retaining observed content and suppressing free space. (c) Typed decoders produce Gaussian components. A check in the original views precedes scene reuse; camera and actor states then control composition and rendering. Scene depictions are schematic, not experimental outputs.

Our contributions are:

- We introduce DriveCompose, a generative reconstruction framework that jointly completes background and canonical actor assets from driving observations. The resulting compositional Gaussian scene can be rendered under new camera and actor states using the same completed content.
- We develop a structured conditional flow model that shares scene context across background and assets through cross-component memory. Observed-code and free-space constraints guide latent completion, while input-view checks validate the assembled scene against the recorded observations.

Section 2 reviews related work, Section 3 presents the method, and Section 4 reports experiments. Sections 5 and 6 discuss limitations and conclude.

2. Related work

Driving-scene reconstruction. NeRF and Gaussian splatting provide the radiance representations underlying urban reconstruction [15, 22], with driving-specific rendering developed in NeuRAD and SplatAD [11, 33]. Feed-forward methods learn geometry and Gaussian prediction from images [2, 4, 12, 35, 36]. For driving logs, STORM aggregates dynamic Gaussians through scene flow, while DGGT combines camera recovery with temporal visibility and motion-guided interpolation [3, 45]. DrivingForward, StreetForward, PointForward, and Instant NuRec further explore surround-view inputs, temporal processing, point-aligned representations, and simulation-oriented reconstruction [6, 14, 32, 48]. TokenGS also decouples Gaussian prediction from pixels through learnable tokens [29]. DriveCompose aggregates recovered evidence onto background and canonical asset supports, then conditionally generates their missing content before Gaussian decoding.

Compositional scenes and controllable assets. Neural Scene Graphs, Street Gaussians, and DrivingGaussian separate scene content into background and moving components [23, 43, 50]. OmniRe includes canonical representations for rigid and articulated actors, while UniSim and HUGSIM connect reconstructed content to closed-loop simulation [5, 46, 49]. At the asset level, SMPL-based body models and animatable Gaussian methods provide canonical surfaces and motion bindings [13, 16, 20, 25, 27]; LHM and IDOL infer animatable humans, and UrbanCAD targets controllable vehicles [21, 28, 52]. These works establish the representation and control mechanisms used to manipulate scene components. Our completion variables retain this component structure, so generated surfaces inherit an asset identity, coordinate frame, and motion binding.

Generative reconstruction and completion. Image-conditioned 3D generation uses generated views [9, 19, 30] or structured latents encoding geometry and appearance [41, 42]. GenAssets learns driving-asset priors, CompleteSplat completes hidden Gaussian content, and JRM aggregates unaligned object observations through latent flow matching [17, 40, 47]. LatentWorld organizes semantic occupancy generation around entity and background latents [24], making structured scene generation an important point of comparison. DriveCompose targets RGB-renderable reconstruction conditioned on driving logs: a shared completion model exchanges context

across world-space background and canonical vehicle and human codes. Observed codes and known free space constrain sampling, and original-view rendering checks the assembled result before reuse.

Generative driving scenes and videos. MagicDrive, DriveDreamer, and Vista generate driving imagery conditioned on scene structure or control [8, 10, 37]. StreetCrafter uses generated views to supervise dynamic Gaussian reconstruction; HorizonForge combines scene editing with video diffusion [38, 44]. WorldSplat generates Gaussians and refines their rendered videos, while DIFIX3D+ applies diffusion-based image refinement [39, 51]. These methods connect generative imagery and three-dimensional scene representations at different stages. DriveCompose performs completion during scene construction and stores the resulting background and canonical assets; subsequent camera, rigid-body, and skeletal states render the same completed content.

3. Method

DriveCompose reconstructs a driving scene in the units that will later be controlled. Figure 1 shows the complete construction and reuse path.

3.1. Problem formulation and compositional representation

The input is a synchronized image log $\mathbf{I}_{t,v}$ with timestamps, camera intrinsics, and camera-to-rig calibration. Rig poses are provided or estimated relative to the first frame. Coordinate conventions and image preprocessing are detailed in Appendix D of the supplementary material.

The output is a compositional Gaussian scene

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-space background, C_{sky} a sky cubemap, V_{sky} its supported directions, $\mathcal{A}_a$ the canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the sampling seed. The components are rendered by Gaussian splatting [15]. Vehicles retain local geometry and dimensions. Humans additionally retain a rest skeleton and skinning weights. Scene state controls the placement and articulation of these fixed components.

We model missing content through conditional generation. Let $\mathcal{Q}$ be the recovered query layout and C its observed evidence. The completion model samples unknown codes $\mathbf{z}_u$, while the codec D_ψ decodes the assembled codes into scene content:

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

The decoder also carries forward the fixed support, identity, binding, and sky-validity metadata in Eq. 1. The layout is established from the input log; generation fills its unknown content. Simulation time changes the state of the decoded scene, while flow time indexes the construction of its latent codes.

3.2. Visual recovery and canonical aggregation

Shared visual encoding. The Driving Log Encoder uses transformer features [7, 34] to predict camera-axis depth, normals, confidence, semantics, and instance assignments from the calibrated image log. Trajectory-level identity slots separate persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Architectural details and state parameterization are given in Appendix D of the supplementary material.

Query supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; vehicle anchors are motion-normalized and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout remains fixed during completion. We obtain $\mathbf{q}_i^w$ by placing the query in the first input configuration, including skinning for humans. Canonical coordinates identify the content within its component; reference-world coordinates let the generator relate that component to its surroundings.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{vehicle}, \\ \text{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection into a camera gives depth $z_{i,t,v}$ and sampled feature, depth, and confidence $(\boldsymbol{\phi}_{i,t,v}, d_{i,t,v}, c_{i,t,v})$. With tolerance $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$, define $r = (d - z)/\eta$. A surface observation must lie inside the image, have valid depth and sufficient confidence, match the query's instance, and satisfy $|r| \le 1$. Background queries additionally require static-background semantics. Let $a_{i,t,v}$ denote this validity test. We aggregate

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

The query confidence is the maximum over accepted observations. A reliable ray votes for free space only when the entire support cell is in front of the measured surface; the criterion is detailed in Appendix D of the supplementary material.

Positive accumulated observation weight marks a query as observed. Otherwise, sufficient free-space votes mark it free, and the remainder is unknown. The masks m_i, f_i, and u_i are mutually exclusive and satisfy $m_i + f_i + u_i = 1$. Sky projection uses camera rotation alone, reliable sky semantics, and zero depth residual; sky queries receive no free-space votes.

3.3. Structured latent representation and decoding

Type-specific encoders map $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$ to codes $\mathbf{z}_i \in \mathbb{R}^D$. Each query keeps its type, ownership, and coordinate frame even though all codes have the same dimension. Visibility masks and confidence are supplied separately to the completion model. Partial and complete observations therefore use the same intrinsic encoding map. Before conditioning, unobserved codes are set to zero; the observed subset is denoted $\mathbf{z}^{\text{obs}}$.

Background, vehicle, and human decoders independently map each query to K child Gaussians confined to its support cell. Human children inherit the query's skinning weights; vehicle children share their asset's rigid transform. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. A separate sky head predicts cubemap colors within the observation-supported domain V_{sky}. Gaussian and appearance parameterizations are detailed in Appendix E of the supplementary material.

Decoding is per query: once the codec is frozen, an unchanged observed code produces unchanged Gaussian parameters. The completion network can exchange context across queries without changing this decoding property.

3.4. Conditional generative reconstruction

Generative completion module. Our generative completion module (GCM) combines a scene-conditioned flow transformer with constrained sampling. We parameterize its completion velocity with a DiT-style transformer [26] trained by flow matching [18]. Its input combines current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and pooled log features. Background groups follow spatial neighborhoods, each asset forms a local group, and sky directions are grouped by cubemap face. Local self-attention processes these groups separately. A cross-component memory (CCM) module then exchanges context through shared memory:

$$\mathbf{M}' = \mathbf{M} + \text{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{6}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \text{Attn}_w(\text{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{7}$$

Here $\widetilde{\mathbf{X}}$ concatenates locally updated tokens, and attention arguments are queries, keys, and values. A residual feed-forward update follows $\mathbf{U}$. Flow-time modulation enters the local normalization; learned tokens and log summaries initialize memory. This structure keeps each support's local organization while allowing a vehicle or human query to use information from the surrounding scene.

Flow training and constrained sampling. Let $\mathbf{z}_1$ be a complete target on the same query layout and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be Gaussian noise. We interpolate only the unknown content at flow time $\tau \in [0, 1]$:

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\text{obs}} + u \odot [(1 - \tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{8}$$

Observed codes remain fixed and free-space codes remain zero. The target velocity for unknown queries is $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i, the objective is

$$\mathcal{L}_{\text{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{9}$$

All unknown queries interpolated in Eq. 8 require valid target codes, even if s_i excludes some of them from the loss. This requirement determines how complete targets are constructed in Sec. 3.6.

At inference, we initialize unknown codes with noise and take S Euler steps with $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f} \left[\mathbf{z}^j + \Delta\tau \, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C) \right]. \tag{10}$$

The projection $\Pi_{m,f}$ restores observed codes and clears free-space codes after every step. The seed fixes one sampled completion. Sampling precedes decoding, and the resulting scene is reused when the camera or actor states change.

3.5. Scene composition and observation consistency

State-driven rendering. At online step k, state s_k specifies the ego pose, active assets, vehicle transforms, and human root and joint poses. Vehicles are placed by rigid transforms and humans by skinning; standard transformations and backend details are given in Appendix C of the supplementary material. The renderer composites the background, active assets, and valid sky:

Table 1. Training schedule. R: recovery; C: codec; G: GCM. T: trained; F: frozen; –: unused.

Stage	R	C	G	Supervision
Recovery	T	–	–	Geometry, semantics, identity, states
Codec	F	T	–	Partial / complete observations
Prior	F	F	T	Complete targets
Adaptation	F	F	T	Real-log rendering and scene constraints

$$o_k = \mathcal{R}(\text{Compose}(\mathcal{W}, s_k), \mathcal{C}_k). \tag{11}$$

Resetting or branching a trajectory changes state while preserving $\mathcal{W}$ and its seed. Finite support and supported sky directions determine coverage.

Checking a constructed scene. Fixed observed codes preserve decoded parameters, but a newly generated surface can still occlude them. Before accepting a scene, we render the completed candidate and an observed-only reference at the original cameras and actor states. The reference disables all non-observed queries. Let $e_{\text{cand}}(p)$ and $e_{\text{obs}}(p)$ be their mean absolute RGB errors at pixel p. We measure local degradation with

$$\Delta e(p) = \max\{e_{\text{cand}}(p) - e_{\text{obs}}(p), 0\}. \tag{12}$$

Averaging this quantity over reliable non-ego pixels prevents improvements in one region from cancelling degradation elsewhere. Acceptance also bounds absolute RGB and valid-depth errors, minimum coverage, and foreground intrusion, using aggregate and worst-view checks. An intrusion is a newly rendered surface in front of the predicted reference depth that is absent from the observed-only render; its rate is normalized over reliable non-sky geometry pixels. Predicted depth provides an observation-consistency reference. Independent geometry evaluation requires a separate reference. A rejected candidate is not exported for reuse.

3.6. Training procedure

Learning is divided into four stages so that the completion target has a stable meaning (Table 1).

For codec fitting, partial input and more complete observations share the same query positions, types, order, and ownership. With recovery and the entire codec frozen, complete scene or asset examples provide a valid target for every unknown query to train the GCM with Eq. 9. Real-log adaptation differentiates through flow integration, decoding, and rendering to update only the GCM. Complete-target samples can be interleaved to retain the prior, with each loss applied only to samples carrying its required targets. Observed anchors are enforced by frozen decoding and hard code restoration. Detailed supervision, losses, and conditioning perturbations are provided in Appendix B of the supplementary material.

At inference, a log is encoded once; its support and evidence remain fixed during completion, and the decoded scene is checked before reuse under requested states (Section 3.5).

4. Experiments

We evaluate our method along three axes: *(i)* dynamic scene reconstruction quality on Waymo Open Dataset (§ 4.2); *(ii)* off-trajectory view synthesis under lateral camera shifts (§ 4.3); and *(iii)* ablations of the main design choices (§ 4.4). Additional cross-dataset generalization results are provided in the supplementary material.

4.1. Experimental Setup

Datasets and protocol. We use the Waymo Open Dataset with 798 training and 202 validation sequences. Following STORM, each validation sequence is divided into non-overlapping 2-second clips of 20 frames at 10 Hz [45]. The FRONT, FRONT_LEFT, and FRONT_RIGHT cameras provide observations at zero-based frame indices $\{0, 5, 10, 15\}$. The remaining 16 frames are used for evaluation: 12 frames for temporal interpolation and four frames for prediction beyond the last observation. Predictions and reference images are resized to 240×160 pixels for evaluation [14].

Baselines. We compare with STORM, DGGT, StreetForward [3, 45, 48]. Methods use the same context timestamps and target views, with model-specific preprocessing. Camera parameters, additional geometric inputs, and rendering adapters are documented with the implementation settings. For spatial extrapolation, we compare Gaussian renderings before diffusion refinement, following PointForward [6].

Metrics. We report PSNR and SSIM over the full image and shared annotated regions. Depth quality is measured using AbsRel and δ_1 at valid projected LiDAR samples after per-frame scale alignment [14]. Scores are averaged over frames within each clip and then over clips within each sequence; sequences receive equal weight. Novel-trajectory renderings are evaluated using FID and FVD against a fixed real-image and video reference population.

Implementation details. DriveCompose reconstructs category-specific actor assets from motion-aligned observations and composes them with a static background.

Reconstruction time includes preprocessing, asset generation, and scene composition. Rendering time is measured separately, and asset reuse is evaluated across repeated scene queries.

4.2. Dynamic Scene Reconstruction

We evaluate reconstruction at unobserved times along the recorded camera trajectory. Table 2 reports full-image, dynamic-region, and depth quality; Fig. 2 compares rendered appearance across five examples.

Quantitative comparison. DriveCompose achieves the best scores across all six metrics in Table 2. Full-image and dynamic-region PSNR exceed the strongest baseline in each category by 1.154 and 2.340 dB, respectively. The corresponding SSIM scores reach 0.901 and 0.862, while depth AbsRel decreases to 0.095 and δ_1 reaches 0.877.

Qualitative comparison. Figure 2 shows that DriveCompose retains the pedestrian silhouette in the first row and the rear profile of the central vehicle in the second row, while preserving the surrounding street layout. The baseline renderings exhibit displaced or repeated actor contours in the upper rows and blurred or distorted road structure in the lower rows. DriveCompose more closely follows these reference structures in the displayed examples, with residual artifacts around the pedestrians in the third row and the parked pickup in the last row.

Table 2. Dynamic scene reconstruction on Waymo. Image quality is reported over the full image and annotated regions; depth is evaluated at valid LiDAR samples.

	Full image		Dynamic Region		Depth	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	26.380	0.826	22.100	0.624	0.178	0.741
DGGT	27.410	0.794	22.710	0.651	0.249	0.668
StreetForward	27.010	0.818	20.919	0.592	0.197	0.759
Ours	**28.564**	**0.901**	**25.050**	**0.862**	**0.095**	**0.877**

4.3. Off-Trajectory View Synthesis

We evaluate off-trajectory rendering under lateral shifts of ± 1, ± 2, and ± 4 meters, keeping camera orientation, intrinsics, and timestamps fixed [6, 51].

Table 3 reports FID and FVD averaged over separately evaluated left and right shifts at each magnitude, using the same real reference population across methods. Figure 3 presents qualitative view-extrapolation results on Waymo with the camera shifted 2 m to the left, illustrating vehicle shape, road layout, and occlusion boundaries across five scenes.

Table 3. Novel-view synthesis under lateral camera shifts. FID and FVD are averaged over left and right shifts at each displacement magnitude. Lower is better.

Method	Shift ± 1 m		Shift ± 2 m		Shift ± 4 m	
	FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
STORM	61.97	722.27	83.72	1029.36	132.57	1704.16
DGGT	73.99	984.00	93.36	1197.03	128.01	1661.70
StreetForward	84.30	878.95	103.31	1095.02	123.58	1578.63
Ours	**28.52**	**544.27**	**37.43**	**673.17**	**58.19**	**817.13**

4.4. Ablation studies

Protocol. Paired evaluation of hidden surfaces is defined on independent complete-reference scenes; Waymo logs provide recorded-view evaluation. Each 29-timestamp window uses three synchronized cameras: front, front-left, and front-right. The nested input sets are $\mathcal{T}_2 = \{0, 28\}$, $\mathcal{T}_4 = \{0, 8, 20, 28\}$, and $\mathcal{T}_8 = \{0, 4, \ldots, 28\}$, with shared target timestamps $\{2, 6, \ldots, 26\}$. Target images are excluded from evidence and query construction; target camera and actor states are supplied equally to every variant. The recovery module and codec are frozen, with matched query layouts, data exposure, and optimization budgets at each input setting. Training samples $T \in \{2, 4, 8\}$ uniformly; completion uses 24 Euler steps. The two learned variants use training seeds $\{0, 1, 2\}$ and world seeds $\{7, 17, 27\}$; the variant w/o GCM is deterministic. Hyperparameters and acceptance thresholds are selected on an internal development split before evaluation on scene- and identity-disjoint data.

Metrics. New-MAE measures linear-RGB error on independently identified surfaces unseen in every input; NVS-MAE covers all held-out target pixels. Both MAE metrics and Obs-Reg use linear RGB on a $[0, 1]$ scale. Across input settings, New-MAE uses the common region unseen even under $\mathcal{T}_8$. Obs-Reg averages the positive input-view regression in Eq. 12. Cov. is the non-sky target fraction with support coverage > 0.5, opacity > 0.1, and finite positive depth. Uncovered pixels receive the same zero-RGB fill for error measurement. Pass is the fraction of all requested worlds accepted by the fixed checker, including construction failures in its denominator. Image metrics include every renderable candidate, including rejected worlds; non-renderable counts are recorded separately. Metrics are averaged within each scene and then across scenes and training seeds.

Completion and scene context. Table 4 compares the full model with two controls. The variant w/o GCM bypasses the generative completion module and disables the opacity of every non-observed, non-sky query. The variant w/o CCM



Figure 2. **Qualitative comparison for dynamic scene reconstruction.** Each row shows one example. Columns, from left to right, are ground truth, DGGT, STORM, StreetForward, and DriveCompose (Ours). The examples cover pedestrians, traffic, and urban backgrounds, allowing comparison of actor silhouettes, vehicle boundaries, and road and building structure.

replaces the cross-component memory module with isolated per-component memories at every layer, while sharing parameters and retaining the same global log context. The learned control is retrained through Stages 3–4; memory-related compute differences are recorded.

Removing completion raises New-MAE from 0.062 to 0.186 and reduces coverage from 96.3% to 74.6%. The variant w/o GCM has a 95.0% pass rate, but leaves substantially more target pixels uncovered. CCM reduces New-MAE from 0.078 (w/o CCM) to 0.062 (Ours), a 20.5% relative reduction, while increasing coverage by 1.8 percentage points and the pass rate by 6.0 points.

Input sparsity. Table 5 reuses checkpoints of the full model and the variant w/o CCM at $T = 2, 4, 8$, with 6/12/24 input images. The time span, target views, and common unseen-region mask remain fixed; evidence and queries are rebuilt from each input subset. No model is retrained for this comparison. For the full model, increasing T from 2 to 8 reduces NVS-MAE from 0.043 to 0.026 and New-MAE from 0.092 to 0.062, while increasing coverage from 91.7% to 96.3%. The full model outperforms the variant w/o CCM on all three metrics at every input count. The absolute New-MAE reduction from CCM is 0.027, 0.020, and 0.016 at $T = 2, 4, 8$, respectively, while the correspond-

Table 4. **Component ablations.** All variants use $T = 8$ (24 images), shared recovery and codec, and common evaluation masks. Quality is measured before acceptance filtering; Pass includes all requested worlds. A dash denotes an unreported metric; bold marks the best reported value in each column.

Variant	New-MAE ↓	Obs-Reg ↓	Cov. (%) ↑	Pass (%) ↑
w/o GCM	0.186	–	74.6	**95.0**
w/o CCM	0.078	0.005	94.5	87.0
Ours	**0.062**	**0.003**	**96.3**	93.0



Figure 3. **Qualitative view extrapolation on Waymo with a 2 m leftward camera shift.** Each column shows one scene. Rows, from top to bottom, show the reference image (GT), STORM, DGGT, and DriveCompose (Ours).

Table 5. **Input-timestamp sensitivity.** At each T, we compare w/o CCM with the full model using fixed checkpoints and target views. New-MAE uses the common unseen region. Results at $T = 8$ match Table 4; bold marks the better value within each group.

T	Variant	NVS-MAE ↓	New-MAE ↓	Cov. (%) ↑
2	w/o CCM	0.052	0.119	88.5
	Ours	**0.043**	**0.092**	**91.7**
4	w/o CCM	0.038	0.094	92.3
	Ours	**0.032**	**0.074**	**94.6**
8	w/o CCM	0.030	0.078	94.5
	Ours	**0.026**	**0.062**	**96.3**

ing coverage gains are 3.2, 2.3, and 1.8 percentage points. The larger absolute gains at lower input counts indicate that cross-component context is especially useful when observations are sparse.

4.5. State-driven reuse

At timestamp 14 and world seed 7, the interventions independently translate the camera by 1 m along its local $+x$ axis, rotate a vehicle by 30° about its root and the world vertical axis, or apply a held-out valid human pose H_1 with fixed root placement. Lighting and canonical assets remain fixed, without resampling completion.

5. Limitations

DriveCompose represents a finite region under fixed illumination. Errors in recovered depth, identity, or canonical pose can enter the observed anchors and constrain later completion incorrectly. Vehicles are rigid, and humans retain fixed clothing volume and skeleton binding; poses outside the template's support can expose missing geometry. Reference-world coordinates provide context at construction, while contact and collision validity after an edit depend on the physical backend. The observation check assesses agreement with recorded evidence. Geometry outside that evidence and appearance under changed illumination require separate supervision or modeling.

6. Conclusion

We presented DriveCompose, a generative reconstruction framework for compositional driving scenes. Its structured latent representation gathers image evidence into background and canonical assets, giving missing content a defined location, identity, and motion model before generation. Conditional completion uses scene context while retaining observed evidence, and typed decoding produces the components of a reusable Gaussian scene. Separating content construction from state-driven composition allows subsequent camera and actor changes to reuse the same completion.

References

[1] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuScenes: A Multimodal Dataset for Autonomous Driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11621–11631, 2020. 1

[2] David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelSplat: 3D Gaussian Splats from Image Pairs for Scalable Generalizable 3D Reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19457–19467, 2024. 2

[3] Xiaoxue Chen, Ziyi Xiong, Yuantao Chen, Gen Li, Nan Wang, Hongcheng Luo, Long Chen, Haiyang Sun, Bing Wang, Guang Chen, Hongyang Li, Ya-Qin Zhang, Hangjun Ye, and Hao Zhao. DGGT: Feedforward 4D Reconstruction of Dynamic Driving Scenes using Unposed Images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1265–1276, 2026. 1, 2, 5

[4] Yuedong Chen, Haofei Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. MVSplat: Efficient 3D Gaussian Splatting from Sparse Multi-View Images. In *European Conference on Computer Vision (ECCV)*, 2024. 2

[5] Ziyu Chen, Jiawei Yang, Jiahui Huang, Riccardo de Lutio, Janick Martinez Esturo, Boris Ivanovic, Or Litany, Zan Gojcic, Sanja Fidler, Marco Pavone, Li Song, and Yue Wang. OmniRe: Omni Urban Scene Reconstruction. In *International Conference on Learning Representations*, pages 85508–85527, 2025. 1, 2

[6] Cheng Chi, Xianqi Wang, Hongcheng Luo, Mingfei Tu, Gangwei Xu, Zehan Zhang, Bing Wang, Guang Chen, Hangjun Ye, Sida Peng, Xin Yang, and Haiyang Sun. PointForward: Feedforward Driving Reconstruction through Point-Aligned Representations. *arXiv preprint arXiv:2605.11594*, 2026. 1, 2, 5, 6

[7] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*, 2021. 3

[8] Ruiyuan Gao, Kai Chen, Enze Xie, Lanqing HONG, Zhenguo Li, Dit-Yan Yeung, and Qiang Xu. MagicDrive: Street View Generation with Diverse 3D Geometry Control. In *International Conference on Learning Representations*, pages 22841–22860, 2024. 3

[9] Ruiqi Gao, Aleksander Hołyński, Philipp Henzler, Arthur Brussee, Ricardo Martin-Brualla, Pratul Srinivasan, Jonathan T. Barron, and Ben Poole. CAT3D: Create Anything in 3D with Multi-View Diffusion Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 75468–75494, 2024. 2

[10] Shenyuan Gao, Jiazhi Yang, Li Chen, Kashyap Chitta, Yihang Qiu, Andreas Geiger, Jun Zhang, and Hongyang Li. Vista: A Generalizable Driving World Model with High Fidelity and Versatile Controllability. In *Advances in Neural Information Processing Systems*, pages 91560–91596, 2024. 3

[11] Georg Hess, Carl Lindström, Maryam Fatemi, Christoffer Petersson, and Lennart Svensson. SplatAD: Real-Time Lidar and Camera Rendering with 3D Gaussian Splatting for Autonomous Driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11982–11992, 2025. 2

[12] Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. LRM: Large Reconstruction Model for Single Image to 3D. In *International Conference on Learning Representations*, pages 50678–50702, 2024. 2

[13] Shoukang Hu, Tao Hu, and Ziwei Liu. GauHuman: Articulated Gaussian Splatting from Monocular Human Videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20418–20431, 2024. 2

[14] Jiahui Huang, Jiawei Ren, Michal Tyszkiewicz, Bjoern Haefner, Michael Shelley, Xin Kang, Seung Wook Kim, Ning Xu, Qi Wu, Janick Martinez Esturo, Shengyu Huang, Nick Schneider, Laura Leal-Taixé, Zan Gojcic, and Sanja Fidler. Instant NuRec: Feed-Forward 3D Gaussian Reconstruction for Driving Scene Simulation. *arXiv preprint arXiv:2607.14203*, 2026. 2, 5

[15] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics*, 42(4), 2023. 2, 3

[16] Muhammed Kocabas, Jen-Hao Rick Chang, James Gabriel, Oncel Tuzel, and Anurag Ranjan. HUGS: Human Gaussian Splats. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 505–515, 2024. 2

[17] Ziwei Liao, Mohamed Sayed, Steven L. Waslander, Sara Vicente, Daniyar Turmukhambetov, and Michael Firman. Complete Gaussian Splats from a Single Image with Denoising Diffusion Models. *arXiv preprint arXiv:2508.21542*, 2025. 1, 2

[18] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow Matching for Generative Modeling. In *International Conference on Learning Representations*, 2023. 1, 4

[19] Ruoshi Liu, Rundi Wu, Basile Van Hoorick, Pavel Tokmakov, Sergey Zakharov, and Carl Vondrick. Zero-1-to-3: Zero-shot One Image to 3D Object. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9298–9309, 2023. 2

[20] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A Skinned Multi-Person Linear Model. *ACM Trans. Graphics (Proc. SIGGRAPH Asia)*, 34(6):248:1–248:16, 2015. 2

[21] Yichong Lu, Yichi Cai, Shangzhan Zhang, Hongyu Zhou, Haoji Hu, Huimin Yu, Andreas Geiger, and Yiyi Liao. Ur-

banCAD: Towards Highly Controllable and Photorealistic 3D Vehicles for Urban Scene Simulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 27519–27530, 2025. 2

[22] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *European Conference on Computer Vision (ECCV)*, 2020. 2

[23] Julian Ost, Fahim Mannan, Nils Thuerey, Julian Knodt, and Felix Heide. Neural Scene Graphs for Dynamic Scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2856–2865, 2021. 2

[24] Jinhyung Park, Navyata Sanghvi, Erica Weng, Shawn Hunt, Shinya Tanaka, Hironobu Fujiyoshi, and Kris Kitani. Grounded Latents for Entity-Centric 4D Scene Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2

[25] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. Expressive Body Capture: 3D Hands, Face, and Body From a Single Image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10975–10985, 2019. 2

[26] William Peebles and Saining Xie. Scalable Diffusion Models with Transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4195–4205, 2023. 4

[27] Zhiyin Qian, Shaofei Wang, Marko Mihajlovic, Andreas Geiger, and Siyu Tang. 3DGS-Avatar: Animatable Avatars via Deformable 3D Gaussian Splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5020–5030, 2024. 2

[28] Lingteng Qiu, Xiaodong Gu, Peihao Li, Qi Zuo, Weichao Shen, Junfei Zhang, Kejie Qiu, Weihao Yuan, Guanying Chen, Zilong Dong, and Liefeng Bo. LHM: Large Animatable Human Reconstruction Model for Single Image to 3D in Seconds. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14184–14194, 2025. 2

[29] Jiawei Ren, Michal Jan Tyszkiewicz, Jiahui Huang, and Zan Gojcic. TokenGS: Decoupling 3D Gaussian Prediction from Pixels with Learnable Tokens. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15365–15375, 2026. 2

[30] Yichun Shi, Peng Wang, Jianglong Ye, Long Mai, Kejie Li, and Xiao Yang. MVDream: Multi-view Diffusion for 3D Generation. In *International Conference on Learning Representations*, pages 39838–39859, 2024. 2

[31] Pei Sun, Henrik Kretzschmar, Xerxes Dotiwalla, Aurelien Chouard, Vijaysai Patnaik, Paul Tsui, James Guo, Yin Zhou, Yuning Chai, Benjamin Caine, Vijay Vasudevan, Wei Han, Jiquan Ngiam, Hang Zhao, Aleksei Timofeev, Scott Ettinger, Maxim Krivokon, Amy Gao, Aditya Joshi, Yu Zhang, Jonathon Shlens, Zhifeng Chen, and Dragomir Anguelov. Scalability in Perception for Autonomous Driving: Waymo Open Dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2446–2454, 2020. 1

[32] Qijian Tian, Xin Tan, Yuan Xie, and Lizhuang Ma. DrivingForward: Feed-forward 3D Gaussian Splatting for Driving Scene Reconstruction from Flexible Surround-view Input. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 7374–7382, 2025. 2

[33] Adam Tonderski, Carl Lindström, Georg Hess, William Ljungbergh, Lennart Svensson, and Christoffer Petersson. NeuRAD: Neural Rendering for Autonomous Driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14895–14904, 2024. 2

[34] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is All you Need. In *Advances in Neural Information Processing Systems*, 2017. 3

[35] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual Geometry Grounded Transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5294–5306, 2025. 2

[36] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. DUSt3R: Geometric 3D Vision Made Easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20697–20709, 2024. 2

[37] Xiaofeng Wang, Zheng Zhu, Guan Huang, Chen Xinze, Jiagang Zhu, and Jiwen Lu. DriveDreamer: Towards Real-world-driven World Models for Autonomous Driving. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[38] Yifan Wang, Francesco Pittaluga, Zaid Tasneem, Chenyu You, Manmohan Chandraker, and Ziyu Jiang. HorizonForge: Driving Scene Editing with Any Trajectories and Any Vehicles. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 3

[39] Jay Zhangjie Wu, Yuxuan Zhang, Haithem Turki, Xuanchi Ren, Jun Gao, Mike Zheng Shou, Sanja Fidler, Zan Gojcic, and Huan Ling. DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26024–26035, 2025. 3

[40] Qirui Wu, Yawar Siddiqui, Duncan Frost, Samir Aroudj, Armen Avetisyan, Richard Newcombe, Angel X. Chang, Jakob Engel, and Henry Howard-Jenkins. JRM: Joint Reconstruction Model for Multiple Objects without Alignment. *arXiv preprint arXiv:2603.25985*, 2026. 2

[41] Jianfeng Xiang, Zelong Lv, Sicheng Xu, Yu Deng, Ruicheng Wang, Bowen Zhang, Dong Chen, Xin Tong, and Jiaolong Yang. Structured 3D Latents for Scalable and Versatile 3D Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21469–21480, 2025. 2

[42] Jianfeng Xiang, Xiaoxue Chen, Sicheng Xu, Ruicheng Wang, Zelong Lv, Yu Deng, Hongyuan Zhu, Yue Dong, Hao Zhao, Nicholas Jing Yuan, and Jiaolong Yang. Native and Compact Structured Latents for 3D Generation. In *Proceedings*

of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 14419–14429, 2026. 2

[43] Yunzhi Yan, Haotong Lin, Chenxu Zhou, Weijie Wang, Haiyang Sun, Kun Zhan, Xianpeng Lang, Xiaowei Zhou, and Sida Peng. Street Gaussians: Modeling Dynamic Urban Scenes with Gaussian Splatting. In *European Conference on Computer Vision (ECCV)*, 2024. 1, 2

[44] Yunzhi Yan, Zhen Xu, Haotong Lin, Haian Jin, Haoyu Guo, Yida Wang, Kun Zhan, Xianpeng Lang, Hujun Bao, Xiaowei Zhou, and Sida Peng. StreetCrafter: Street View Synthesis with Controllable Video Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 822–832, 2025. 1, 3

[45] Jiawei Yang, Jiahui Huang, Boris Ivanovic, Yuxiao Chen, Yan Wang, Boyi Li, Yurong You, Apoorva Sharma, Maximilian Igl, Peter Karkus, Danfei Xu, Yue Wang, and Marco Pavone. STORM: Spatio-TempOral Reconstruction Model For Large-Scale Outdoor Scenes. In *International Conference on Learning Representations*, pages 50446–50465, 2025. 1, 2, 5

[46] Ze Yang, Yun Chen, Jingkang Wang, Sivabalan Manivasagam, Wei-Chiu Ma, Anqi Joyce Yang, and Raquel Urtasun. UniSim: A Neural Closed-Loop Sensor Simulator. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1389–1399, 2023. 2

[47] Ze Yang, Jingkang Wang, Haowei Zhang, Sivabalan Manivasagam, Yun Chen, and Raquel Urtasun. GenAssets: Generating in-the-wild 3D Assets in Latent Space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22392–22403, 2025. 1, 2

[48] Zhongrui Yu, Zhao Wang, Yijia Xie, Yida Wang, Xueyang Zhang, Yifei Zhan, and Kun Zhan. StreetForward: Perceiving Dynamic Street with Feedforward Causal Attention. *arXiv preprint arXiv:2603.19552*, 2026. 1, 2, 5

[49] Hongyu Zhou, Longzhong Lin, Jiabao Wang, Yichong Lu, Dongfeng Bai, Bingbing Liu, Yue Wang, Andreas Geiger, and Yiyi Liao. HUGSIM: A Real-Time, Photo-Realistic and Closed-Loop Simulator for Autonomous Driving. *arXiv preprint arXiv:2412.01718*, 2024. 1, 2

[50] Xiaoyu Zhou, Zhiwei Lin, Xiaojun Shan, Yongtao Wang, Deqing Sun, and Ming-Hsuan Yang. DrivingGaussian: Composite Gaussian Splatting for Surrounding Dynamic Autonomous Driving Scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21634–21643, 2024. 2

[51] Ziyue Zhu, Zhanqian Wu, Zhenxin Zhu, Lijun Zhou, Haiyang Sun, Bing Wan, Kun Ma, Guang Chen, Hangjun Ye, Jin Xie, and Jian Yang. WorldSplat: Gaussian-Centric Feed-Forward 4D Scene Generation for Autonomous Driving. *arXiv preprint arXiv:2509.23402*, 2025. 1, 3, 6

[52] Yiyu Zhuang, Jiaxi Lv, Hao Wen, Qing Shuai, Ailing Zeng, Hao Zhu, Shifeng Chen, Yujiu Yang, Xun Cao, and Wei Liu. IDOL: Instant Photorealistic 3D Human Creation from a Single Image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26308–26319, 2025. 2